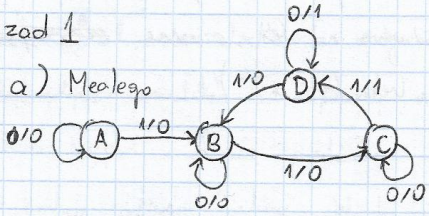
[redacted]

zad 1

a) Mealego



A - 00
B - 01
C - 10
D - 11

x	$Q_A Q_B$	y	$Q_A^+ Q_B^+$	J_a	K_a	J_b	K_b
0	00	0	00	0	0	0	0
0	01	0	01	0	0	0	0
0	10	0	10	0	0	0	0
0	11	1	11	0	0	0	0
1	00	0	01	0	0	1	1
1	01	0	10	1	1	1	1
1	10	1	11	0	0	1	1
1	11	0	01	1	1	0	0

x \ $Q_A Q_B$	00	01	11	10
0	0	0	1	0
1	0	0	0	1

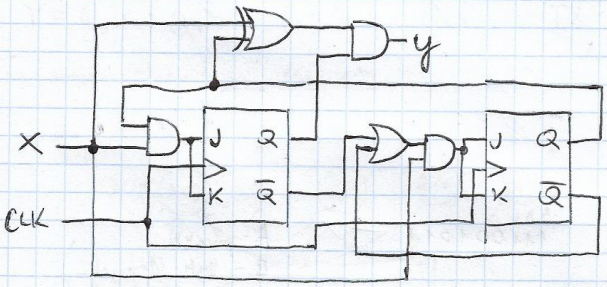
$y = \bar{x} Q_A Q_B + x Q_A \bar{Q_B} =$
$= Q_A (x \oplus Q_B)$

x \ $Q_A Q_B$	00	01	11	10
0	0	0	0	0
1	0	1	1	0

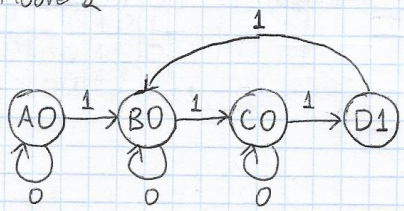
$J_a = K_a = x Q_B$

x \ $Q_A Q_B$	00	01	11	10
0	0	0	0	0
1	1	1	0	1

$J_b = K_b = x (\bar{Q_A} + \bar{Q_B})$



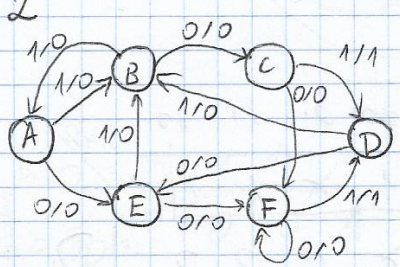
b) Moore'a

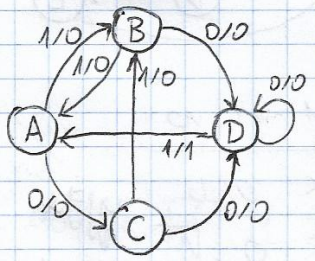


A - 00
B - 01
C - 10
D - 11

Tabela prejść, minimalne funkcje i schemat są identyczne dla tego automatu Moore'a.

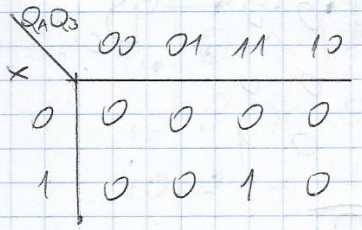
zad 2

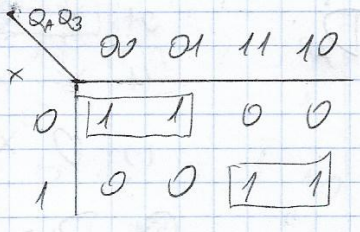


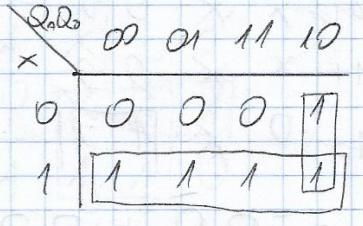


A - 00
B - 01
C - 10
D - 11

x	Q_A	Q_B	y	Q_A^+	Q_B^+	T_A	T_B
0	0	0	0	1	0	1	0
0	0	1	0	1	1	1	0
0	1	0	0	1	1	0	1
0	1	1	0	1	1	0	0
1	0	0	0	0	1	0	1
1	0	1	0	0	0	0	1
1	1	0	0	0	1	1	1
1	1	1	1	0	0	1	1



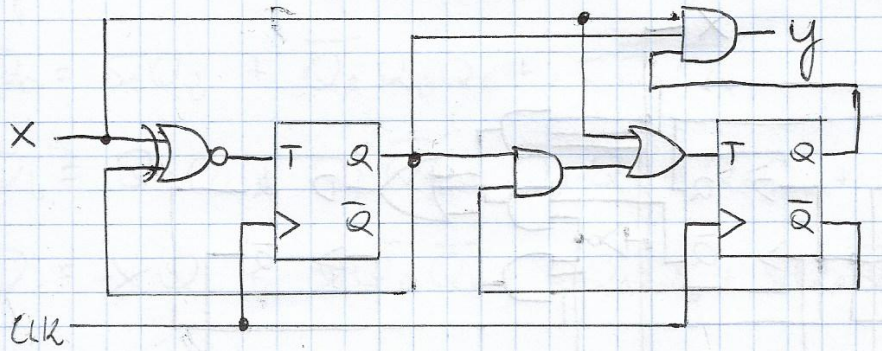




$y = xQ_AQ_B$

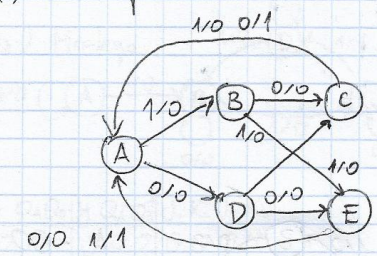
$T_a = \bar{x}\bar{Q}_A + xQ_A = \overline{x \oplus Q_A}$

$T_b = x + Q_A\bar{Q}_B$



zad 3

a) Mealy'ego



E - 111

A - 000

B - 010

C - 110

D - 011

$Q_B Q_C$ / xQ_A	00	01	11	10
00	0	x	0	0
01	x	x	0	1
11	x	x	1	0
10	0	x	0	0

y

$Q_B Q_C$ / xQ_A	00	01	11	10
00	1	x	1	1
01	x	x	0	0
11	x	x	0	0
10	1	x	1	1

D_b

$Q_B Q_C$ / xQ_A	00	01	11	10
00	0	x	1	1
01	x	x	0	0
11	x	x	0	0
10	0	x	1	1

D_A

$Q_B Q_C$ / xQ_A	00	01	11	10
00	1	x	1	0
01	x	x	0	0
11	x	x	0	0
10	0	x	0	1

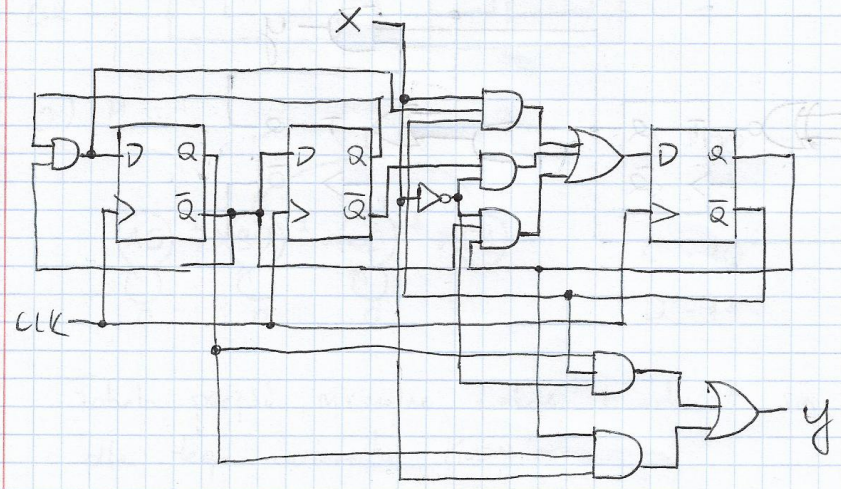
D_C

$y = \bar{x} Q_A \bar{Q_C} + x Q_A Q_C$

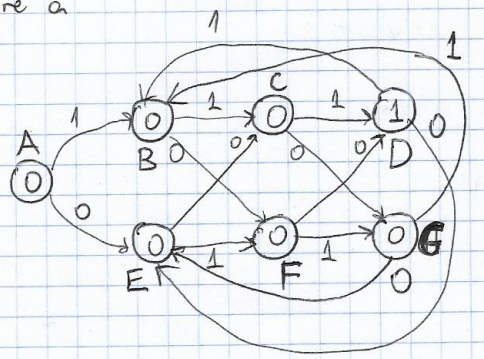
$D_A = \bar{Q_A} Q_B$

$D_b = \bar{Q_A}$

$D_C = \bar{x}\bar{Q_B} + \bar{x}\bar{Q_A}Q_C + x\bar{Q_A}Q_B\bar{Q_C}$



b) Moore'a



A	000
B	001
C	011
D	111
E	100
F	010
G	110

y =

$Q_B Q_C$ / xQ_A	00	01	11	10
00	0	0	0	1
01	0	×	0	0
11	0	×	0	0
10	0	0	1	0

D_a =

$Q_B Q_C$ / xQ_A	00	01	11	10
00	1	0	1	1
01	0	×	1	1
11	0	×	0	0
10	0	0	1	0

D_b =

$Q_B Q_C$ / xQ_A	00	01	11	10
00	0	1	1	1
01	1	×	0	0
11	1	×	0	0
10	0	1	1	1

D_c =

$Q_B Q_C$ / xQ_A	00	01	11	10
00	0	0	0	1
01	1	×	0	0
11	0	×	1	1
10	1	1	1	0

$$y = x\overline{Q}_A Q_B Q_C + \bar{x}\overline{Q}_A Q_B \overline{Q}_C = \overline{Q}_A Q_B (x + \overline{Q}_C)$$

$$D_a = \bar{x} Q_B + \overline{Q}_A Q_B Q_C + \bar{x}\overline{Q}_A \overline{Q}_C$$

$$D_b = \overline{Q}_B Q_C + Q_A \overline{Q}_B + \overline{Q}_A Q_B = \overline{Q}_B Q_C + Q_A \oplus Q_B$$

$$D_c = x Q_C + x\overline{Q}_A \overline{Q}_B + x Q_A Q_B + \bar{x} Q_A \overline{Q}_B + \bar{x}\overline{Q}_A Q_B \overline{Q}_C$$